Monistic Continuum Model (MCM): V08 (EN)

A Monistic Continuum Ontology of Space, Matter and Propagation

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1 Introduction

The Monistic Continuum Model (MCM) is a monistic ontological framework in which a single continuum constitutes the sole primitive reality.

The continuum, denoted by

$$D,$$

is the only ontologically fundamental entity.

Space, matter, geometry, curvature, and all observable structures possess no independent ontology.

They emerge as higher-order organizational states of the continuum.

The purpose of the MCM is therefore not to postulate additional fundamental entities, but to derive all observable structures from the internal organization of a single underlying medium.

Within the MCM, stress, rotation, curvature and updating processes represent organizational descriptions of the continuum itself.

Version V08 extends the V07 formulation through three conceptual developments:

1. The introduction of a characteristic bound length associated with stable vortex structures,

   $$L_V = 2\pi r_u,$$

   providing an explicit geometric scale for bound continuum organization.

2. The introduction of a global stress state

   $$\Sigma_G,$$

   describing the global organization of the continuum.

3. A conceptual separation between local vortex stability and global propagation properties.

The resulting hierarchy may be summarized as

$$D \to \Sigma_G \to \sigma_{AB} \to \gamma_{AB} \to R_{ABCD} \to V \to L_V \to \text{Space} \to \text{Matter}.$$

In this framework, local structures are interpreted as localized manifestations of a globally organized continuum rather than as independent ontological entities.

Warum diese Version stärker ist

Sie macht von Anfang an explizit:

MCM ist eine Ontologie, nicht primär eine Physiktheorie. DD ist die einzige ontologische Primitive. Raum und Materie besitzen keine unabhängige Ontologie. Σ_G wird als Organisationszustand und nicht als zweite Substanz eingeführt. Die komplette Emergenzhierarchie der V08 erscheint bereits in der Einleitung.

Aus Sicht einer Fakultät für Wissenschaftstheorie ist diese Fassung deutlich präziser als die aktuelle Version.

2 Axiom 1: The Medium

The medium is the sole ontological primitive of the Monistic Continuum Model.

It is denoted by

$$D. \tag{1}$$

The medium possesses neither spatial nor temporal structure.

It is not to be interpreted as

- space,
- time,
- matter,
- energy,
- or a classical physical fluid.

The medium represents a continuously existing organizational substrate capable of adopting different states of internal organization.

Observable structures emerge from differences in organization within the medium and not from the existence of additional ontological entities.

Space, matter, geometry, curvature, rotation and all other observable phenomena are therefore interpreted as organizational manifestations of

$$D.$$

No observable structure possesses an ontology independent of the medium.

All observable entities remain fully contained within the ontology of

$$D.$$

The medium therefore constitutes the unique ontological foundation of the model.

Ontology: Only the medium is fundamental.

3 Axiom 2: Pre-Spatial State Map

The state of the medium is represented by

$$C(\alpha, \Omega) = (\rho, v_A, \sigma_{AB}, \tau)\,. \tag{2}$$

The state map does not introduce additional ontological entities.

The quantities

$$\rho, \qquad v_A, \qquad \sigma_{AB}, \qquad \tau$$

represent organizational descriptions of the medium.

They describe different aspects of continuum organization and possess no ontology independent of

$$D.$$

where

- α denotes a medium index,
- Ω denotes a global organizational update state,
- all quantities are pre-spatial,
- all quantities describe states of organization within the medium.

The quantity

$$\Omega$$

does not represent time.
Instead,

$$\Omega$$

labels successive global states of continuum organization.

The state map therefore represents the organizational condition of the continuum prior to the emergence of space, geometry and matter.

Observable structures emerge only from the organizational relations encoded in

$$C(\alpha, \Omega).$$

All quantities contained in the state map remain fully contained within the ontology of

$$D.$$

4 Axiom 3: Stress-Induced Internal Geometry

Internal geometry emerges from the organization of stress within the medium.

The geometric state of the continuum is represented by

$$\gamma_{AB} = \delta_{AB} + \alpha_1 \sigma_{AB} + \alpha_2 I_1 \delta_{AB}, \tag{3}$$

where

$$I_1 = \mathrm{tr}(\sigma). \tag{4}$$

The quantity

$$\gamma_{AB}$$

does not represent an independently existing geometry.
Instead,

$$\gamma_{AB}$$

is an emergent geometric description generated by the organizational state of the medium.

The stress tensor

$$\sigma_{AB}$$

is therefore ontologically prior to the geometric structure

$$\gamma_{AB}.$$

Geometry is not assumed to be fundamental.

Rather, geometry emerges as an organizational manifestation of stress within the medium.

Consequently, neither

$$\sigma_{AB}$$

nor

$$\gamma_{AB}$$

constitutes an independent ontological level.

Both remain fully contained within the ontology of

$$D.$$

The emergence of geometry from stress establishes the first geometric level in the hierarchy of continuum organization.

The ontological relation may therefore be expressed as

$$\sigma_{AB} \rightarrow \gamma_{AB}.$$

This relation indicates an organizational dependency and does not imply the existence of separate ontological entities.

5 Axiom 4: Internal Curvature

Internal curvature emerges from the internal geometry of the medium.

The curvature state is represented by

$$R_{ABCD} = R(\gamma_{AB}). \tag{5}$$

The quantity

$$R_{ABCD}$$

does not constitute a fundamental ontological entity.

Instead,

$$R_{ABCD}$$

is an emergent organizational description derived from the geometric state

$$\gamma_{AB}.$$

The geometric structure

$$\gamma_{AB}$$

is therefore ontologically prior to the curvature state

$$R_{ABCD}.$$

Curvature is not assumed to exist independently of the continuum.

Rather, curvature emerges through the organization of the internal geometry of the medium.

Consequently, neither

$$\gamma_{AB}$$

nor

$$R_{ABCD}$$

introduces an additional ontological level beyond

$$D.$$

Both remain fully contained within the ontology of the medium.

The emergence of curvature from geometry establishes the second geometric level in the hierarchy of continuum organization.

The ontological relation may therefore be expressed as

$$\gamma_{AB} \to R_{ABCD}.$$

This relation represents an organizational dependency within the medium and not a transition between independent ontological entities.

Verbesserungen gegenüber der Originalfassung:

- Clarifies that

  $$R_{ABCD}$$

  is not fundamental.

- Establishes the ontological priority

  $$\gamma_{AB} \to R_{ABCD}.$$

- Prevents curvature from being interpreted as an independent entity.

- Reinforces the monistic ontology of the MCM.

- Maintains complete consistency with Axiom 1, Axiom 2 and Axiom 3.

Diese Version erfüllt vollständig den festgelegten MCM LaTeX Mode.

6 Axiom 5: Rotational Behaviour

Rotation is defined by

$$L_{AB} = \frac{1}{2} \left(\nabla_A v_B - \nabla_B v_A \right). \tag{6}$$

The quantity

$$L_{AB}$$

represents an organizational property of the medium.
It does not describe the rotation of an independent object within a pre-existing space.
Instead,

$$L_{AB}$$

describes a rotational mode of organization arising within the continuum itself.
Rotation is therefore not ontologically independent of

$$D.$$

Rather, it emerges as a specific organizational state of the medium.
The rotational state interacts with the stress organization

$$\sigma_{AB}$$

and the curvature organization

$$R_{ABCD}$$

of the continuum.
Consequently,

$$L_{AB}, \qquad \sigma_{AB}, \qquad R_{ABCD}$$

must not be regarded as separate entities.

They represent mutually related organizational descriptions of the same underlying medium.

The ontological relation may therefore be expressed as

$$\sigma_{AB} \longrightarrow L_{AB} \longrightarrow R_{ABCD},$$

where the arrows denote organizational dependence rather than transitions between independent objects.

Rotation thus constitutes an intermediate level within the hierarchy of continuum organization and plays a fundamental role in the formation of stable bound structures.

Stable rotational organization ultimately contributes to the emergence of bound geometry and vortex structures within the medium.

Ontologischer Gewinn dieser Version:

- Clarifies that

$$L_{AB}$$

  is not the rotation of an object in space.

- Defines rotation as an organizational mode of the continuum itself.
- Preserves the monistic ontology of the MCM.
- Connects rotation explicitly to stress and curvature organization.
- Establishes the role of rotation in the formation of bound geometry and vortex structures.

Diese Version bleibt vollständig im MCM LaTeX Mode und ist frei von Unicode-Mathematikzeichen.

7 Axiom 6: Update Rate

The update rate is represented by

$$\tau \in \mathbb{R}_{\geq 0}. \tag{7}$$

The quantity

$$\tau$$

does not represent physical time.
Nor does it introduce an independent temporal dimension.
Instead,

$$\tau$$

describes the responsiveness of the medium to changes in its organizational state.
It represents an intrinsic measure of continuum updating and belongs entirely to the ontology of

$$D.$$

The update rate is therefore interpreted as an organizational property of the medium rather than as an external temporal parameter.
Observable temporal phenomena do not exist at the level of

$$\tau.$$

They emerge only at higher levels of organization through the evolution of bound geometric structures.
Consequently,

$$\tau$$

is ontologically prior to any emergent notion of time.

The quantity

$$\tau$$

must therefore not be interpreted as clock time, coordinate time, duration, or an independent temporal substance.

It characterizes the capacity of the medium to internally update its organizational state.

The ontological relation may be expressed as

$$D \longrightarrow \tau \longrightarrow DC.$$

This relation indicates that updating is an intrinsic property of the medium and not a process occurring within an already existing temporal framework.

The update rate therefore provides the foundation for continuum evolution while remaining entirely contained within the ontology of

$$D.$$

Warum diese Version stärker ist:

- Clarifies that τ is not time.
- Clarifies that τ is not a hidden temporal dimension.
- Defines τ as a property of continuum organization.
- Establishes the ontological priority of updating over emergent time.
- Preserves the monistic ontology of the MCM.
- Remains fully consistent with Axiom 1 and Axiom 2.

Diese Fassung ist vollständig im MCM LaTeX Mode und frei von Unicode-Problemen.

8 Axiom 7: Internal Updating

The internal evolution of the medium is represented by

$$DC(\alpha, \Omega). \tag{8}$$

The operator

$$D$$

does not represent differentiation with respect to an external time parameter.

Instead,

$$D$$

describes the internal updating activity of the medium.

The quantity

$$DC(\alpha, \Omega)$$

therefore represents the change of the organizational state of the medium between successive update states.

Bound geometry emerges when the organizational state approaches internal stability.

This condition is expressed by

$$DC(\alpha, \Omega) \approx 0. \tag{9}$$

The condition

$$DC(\alpha, \Omega) \approx 0$$

does not imply the absence of organization.

Rather, it denotes a state in which the internal organization of the medium remains persistently self-maintaining.

Stable bound geometry emerges from such locally self-consistent organizational states.

A stable vortex is therefore defined as a persistent bound configuration satisfying

$$DC(\alpha, \Omega) \approx 0.$$

The vortex is not an independent ontological object.

Instead, it represents a stable organizational state of the medium.

Consequently,

$$DC(\alpha, \Omega), \qquad V, \qquad L_V$$

remain fully contained within the ontology of

$$D.$$

The ontological relation may therefore be expressed as

$$D \longrightarrow DC \longrightarrow V.$$

This relation describes organizational dependence within the medium and does not imply the existence of separate ontological entities.

The emergence of stable bound geometry establishes the transition from continuously updating continuum organization to persistent observable structure.

Ontologischer Gewinn dieser Fassung:

- Clarifies that D is an update operator and not a time derivative.
- Clarifies that $DC(\alpha, \Omega)$ describes organizational evolution of the medium.
- Defines the meaning of

  $$DC(\alpha, \Omega) \approx 0$$

  as organizational stability rather than inactivity.
- Defines the vortex as a stable organizational state and not as an independent object.
- Preserves the monistic ontology of the MCM.
- Creates a clean transition from continuum organization to bound geometry.

9 Axiom 8: Constitutive Dynamics

The organizational state of the medium evolves through internal constitutive dynamics.

Stress reorganization is represented by

$$D\sigma_{AB} = F(V_v, \sigma)_{AB}, \tag{10}$$

with

$$V_v(\sigma_{AB}) = v^C \nabla_C \sigma_{AB}. \tag{11}$$

The operator

$$F$$

does not represent an external force law.

Instead,

$$F$$

denotes the internal organizational response of the medium to its current state.

The quantity

$$D\sigma_{AB}$$

therefore describes the reorganization of stress within the continuum itself.

Neither

$$F, \qquad V_v, \qquad \sigma_{AB}$$

introduces independent ontological entities.

They represent different descriptions of the internal organization of the medium.

The constitutive dynamics express the principle that the medium continuously reorganizes its internal structure according to its existing organizational state.

Consequently, stress evolution is interpreted as an intrinsic property of the continuum.

No external agency, force carrier, or independent interaction is required.

The ontological relation may therefore be expressed as

$$D \longrightarrow \sigma_{AB} \longrightarrow D\sigma_{AB}.$$

This relation describes organizational continuity within the medium rather than an interaction between separate entities.

Constitutive dynamics therefore provide the mechanism through which stress, rotation, and curvature remain mutually coupled throughout the evolution of the continuum.

The resulting organizational processes form the basis for the emergence of stable bound geometry and, ultimately, observable structure.

Ontologischer Gewinn dieser Fassung:

- Clarifies that F is not an external force law.
- Defines constitutive dynamics as an intrinsic organizational process of the medium.
- Preserves the monistic ontology of the MCM.

- Prevents stress from being interpreted as an independent physical substance.
- Establishes constitutive dynamics as the link between organization and evolution of the continuum.
- Maintains full consistency with Axioms 1–7.

10 Axiom 9: Global Stress State

The medium possesses a global organizational state represented by

$$\Sigma_G(\Omega). \tag{12}$$

The quantity

$$\Sigma_G$$

does not constitute an independent ontological entity.
Instead,

$$\Sigma_G$$

represents the collective organizational state of the continuum as a whole.

Organization is defined as the degree of coherent structural correlation present within the medium.

Differences in organization give rise to observable differences in stress, geometry, curvature, rotation, space and matter.

The global organizational state is defined by

$$\Sigma_G = \frac{S_G}{S_{G,0}}, \tag{13}$$

where

$$S_G = \int_D \left(a_\sigma \sigma_{AB} \sigma^{AB} + a_R R_{ABCD} R^{ABCD} \right) dV. \tag{14}$$

All local organizational structures remain embedded within the global organizational state.

The inclusion relation is therefore

$$\sigma_V \subseteq \Sigma_G. \tag{15}$$

The quantity

$$\Sigma_G$$

is ontologically prior to all local organizational structures.

The emergence hierarchy may therefore be expressed as

$$D \longrightarrow \Sigma_G \longrightarrow \sigma_V \longrightarrow V.$$

The global organizational state therefore provides the foundation from which all stable vortex structures emerge.

11 Axiom 10: Emergence of Matter

Matter is not fundamental.

Instead, matter emerges as a stable continuum organization generated by bound vortex structures.

Matter is represented by

$$M = \text{stable bound vortex organization.} \tag{16}$$

The quantity

$$M$$

does not constitute an independent ontological substance.

Rather,

$$M$$

represents a persistent organizational state of the continuum.

Different observable particle classes correspond to different stable classes of vortex organization.

The emergence hierarchy may therefore be expressed as

$$D \longrightarrow \Sigma_G \longrightarrow V \longrightarrow M.$$

Matter remains fully contained within the ontology of

$$D.$$

No material structure possesses an ontology independent of the continuum.

Matter therefore represents a physically observable manifestation of stable continuum organization.

The physical correspondence principle is

$$V \longrightarrow M \longrightarrow \text{Observable Physics.}$$

Observable matter is therefore interpreted as the physical realization of stable vortex organization.

12 Axiom 11: Emergence of Space

Space is not fundamental.

Space emerges from stable vortex organization.

The spatial projection is represented by

$$x^i = X^i(\alpha, \Omega). \tag{17}$$

The mapping

$$X^i(\alpha, \Omega)$$

does not create an independent spatial ontology.

Instead, it provides an observational representation of organized continuum states.

Stable bound vortex organization generates persistent correlation relations within the medium.

The spatial projection emerges as a descriptive representation of these stable correlation structures.

The emergence hierarchy is

$$D \longrightarrow V \longrightarrow X^i \longrightarrow \text{Space}. \tag{18}$$

Observable geometry is therefore interpreted as a projection of underlying continuum organization.

Consequently,

$$\text{Space}$$

possesses no ontology independent of

$$D.$$

Space is therefore not the container of organized structures.

Rather, space emerges as the representation of organized structures already present within the continuum.

13 Axiom 12: Global Stress State

The medium possesses a global organizational state represented by

$$\Sigma_G(\Omega). \tag{19}$$

The quantity

$$\Sigma_G$$

does not constitute an independent ontological entity.

Instead,

$$\Sigma_G$$

represents the collective organizational state of the continuum as a whole.

The global organizational state is defined by the normalized stress functional

$$\Sigma_G = \frac{S_G}{S_{G,0}}, \tag{20}$$

where

$$S_G = \int_D \left(a_\sigma \sigma_{AB} \sigma^{AB} + a_R R_{ABCD} R^{ABCD} \right) dV. \tag{21}$$

The term

$$\sigma_{AB}\sigma^{AB}$$

represents the local stress invariant.

The term

$$R_{ABCD}R^{ABCD}$$

represents the local curvature invariant.

The constants

$$a_\sigma$$

and

$$a_R$$

determine the relative contribution of stress and curvature organization to the global state.

The quantity

$$S_{G,0}$$

denotes a reference continuum state.

Consequently,

$$\Sigma_G = 1$$

corresponds to the chosen reference condition.

All local organizational structures remain embedded within the global organizational state.

The inclusion relation is therefore

$$\sigma_V \subseteq \Sigma_G. \tag{22}$$

The quantity

$$\Sigma_G$$

is ontologically prior to all local organizational structures.

Local stress, rotation, curvature and vortex configurations are interpreted as localized manifestations of the global continuum organization.

The emergence hierarchy may therefore be expressed as

$$D \longrightarrow \Sigma_G \longrightarrow \sigma_V \longrightarrow V.$$

The global stress state therefore provides the organizational foundation from which all stable vortex structures emerge.

14 Physical Correspondence

The purpose of physical correspondence is to establish a systematic relationship between continuum organization and observable physical phenomena.

The current correspondence structure of MCM V08 is summarized as follows.

MCM Quantity	Physical Interpretation
D	Continuum substrate
Σ_G	Global organizational state
σ_{AB}	Local stress organization
γ_{AB}	Emergent internal geometry
R_{ABCD}	Emergent curvature organization
L_{AB}	Rotational organization
V	Stable bound vortex organization
L_V	Characteristic bound length scale
$L_V \tau_V = \Lambda$	Local stability condition
Space	Spatial representation of stable organization
M	Stable matter organization
c	Global propagation property
$c = F(\Sigma_G)$	Propagation determined by global organization

The correspondence hierarchy may therefore be expressed as

$$D \longrightarrow \Sigma_G \longrightarrow \begin{cases} c \\ \sigma_{AB} \longrightarrow \gamma_{AB} \longrightarrow R_{ABCD} \longrightarrow L_{AB} \longrightarrow V \longrightarrow M \end{cases}$$

Observable physical phenomena are interpreted as manifestations of stable continuum organization.

No observable quantity introduces an ontological entity independent of

$$D.$$

The current correspondence framework establishes the conceptual bridge between ontology and physics.

A complete quantitative correspondence remains an open problem of the theory.

15 Local Vortex Stability

A stable vortex possesses a characteristic geometric length

$$L_V$$

and a characteristic update rate

$$\tau_V.$$

Neither

$$L_V$$

nor

$$\tau_V$$

constitutes an independent ontological entity.

Both quantities describe properties of an already existing bound organizational state of the medium.

The local stability condition of a vortex is defined by

$$L_V \tau_V = \Lambda, \tag{23}$$

where

$$\Lambda$$

denotes a characteristic stability constant of bound geometry.

The quantity

$$\Lambda$$

is not interpreted as a fundamental substance or object.

Instead, it characterizes the stable organizational relationship between the geometric scale of a vortex and its update behavior.

The stability relation

$$L_V \tau_V = \Lambda$$

describes only the local stability of an individual bound structure.

It does not determine the global organizational state of the medium.

Likewise, it does not define global propagation properties.

The quantities

$$L_V, \qquad \tau_V, \qquad \Lambda$$

remain entirely contained within the ontology of

$$D.$$

Local vortex stability therefore represents a specialized organizational condition of the continuum.

The emergence hierarchy may be expressed as

$$D \longrightarrow \Sigma_G \longrightarrow V \longrightarrow L_V \longrightarrow (L_V \tau_V = \Lambda)\,.$$

This hierarchy establishes the ontological priority of

$$\Sigma_G$$

and

$$V$$

over the local stability relation.

A stable vortex is therefore interpreted as a persistent bound organizational state whose geometric and update properties satisfy the condition

$$L_V \tau_V = \Lambda.$$

The local stability law provides the foundation for the persistence of bound geometry and thus for the formation of stable observable structures within the continuum.

- Clarifies that

$$L_V, \qquad \tau_V, \qquad \Lambda$$

  are organizational quantities and not independent entities.

- Defines the local stability law as a property of bound geometry.
- Separates local stability from global continuum organization.
- Maintains the ontological priority of

$$\Sigma_G.$$

- Prepares the transition to the next section

$$c = F(\Sigma_G),$$

  without mixing local and global levels.

- Remains fully consistent with Axioms 1–12.

16 Global Propagation State

The propagation properties of the continuum are determined by the global organizational state of the medium.

This global state is represented by

$$\Sigma_G.$$

The propagation relation is expressed symbolically by

$$c = F(\Sigma_G). \tag{24}$$

The quantity

$$c$$

is interpreted as a property of the global continuum state.

It does not arise from the radius, geometry, shape, or local deformation of an individual vortex.

Consequently,

$$c$$

is not determined by

$$r_u, \qquad L_V, \qquad \tau_V,$$

or by any other local geometric property taken in isolation.

Instead,

$$c$$

depends upon the global organizational condition of the medium represented by

$$\Sigma_G.$$

The quantity

$$F$$

denotes a presently unspecified mapping between the global organizational state and the propagation properties of the continuum.

Its explicit mathematical form remains an open problem of the theory.

The relation

$$c = F(\Sigma_G)$$

does not introduce a second ontology.

Neither

$$c$$

nor

$$\Sigma_G$$

constitutes an independent substance or entity.

Both quantities remain fully contained within the ontology of

$$D.$$

The introduction of

$$\Sigma_G$$

therefore preserves the monistic character of the MCM.

Local vortex stability and global propagation must be distinguished.

Local stability is characterized by

$$L_V \tau_V = \Lambda,$$

whereas global propagation is characterized by

$$c = F(\Sigma_G).$$

These relations describe different organizational levels of the continuum.

The emergence hierarchy may therefore be expressed as

$$D \longrightarrow \Sigma_G \longrightarrow c$$

and independently

$$D \longrightarrow \Sigma_G \longrightarrow \sigma_{AB} \longrightarrow \gamma_{AB} \longrightarrow R_{ABCD} \longrightarrow L_{AB} \longrightarrow V \longrightarrow L_V.$$

This distinction establishes the ontological priority of global continuum organization over local bound structures.

Observable propagation properties are therefore interpreted as manifestations of the global state of the medium rather than consequences of individual vortex geometry.

- Clarifies that

  $$c$$

  is a global property of the continuum.

- Establishes the ontological priority of

  $$\Sigma_G.$$

- Separates global propagation from local vortex stability.

- Preserves the monistic ontology of the MCM.

- Defines

  $$F(\Sigma_G)$$

  as an open research problem rather than a completed physical law.

- Creates a clean conceptual bridge between the MCM core ontology and future propagation theory.

- Remains fully consistent with Axioms 1–12 and the Local Vortex Stability section.

17 Emergence Hierarchy

The complete organizational hierarchy of MCM V08 is

$$D \longrightarrow \Sigma_G \longrightarrow \sigma_{AB} \longrightarrow \gamma_{AB} \longrightarrow R_{ABCD} \longrightarrow L_{AB} \longrightarrow V \longrightarrow L_V \longrightarrow \text{Space} \longrightarrow \text{Matter}. \tag{25}$$

This hierarchy represents successive organizational levels within a single continuum. All structures remain fully contained within the ontology of

$$D.$$

The global organizational state

$$\Sigma_G$$

provides the foundation for all local organizational structures.
The propagation hierarchy is

$$D \longrightarrow \Sigma_G \longrightarrow c. \tag{26}$$

The matter correspondence hierarchy is

$$D \longrightarrow \Sigma_G \longrightarrow V \longrightarrow M \longrightarrow \text{Observable Physics.} \tag{27}$$

Stage VIII establishes the connection between continuum organization and observable physical phenomena.

Observable physics is interpreted as the manifestation of stable continuum organization and not as an independent ontological level.

No observable quantity possesses an ontology independent of

$$D.$$

18 Interpretation of Matter

Matter corresponds to long-lived stable vortex configurations embedded within the global continuum organization.

Observable particles are interpreted as physically realized forms of stable vortex organization.

Examples include

- proton-like vortex states,
- neutron-like vortex states,
- composite bound vortex systems.

Different particle classes correspond to different stable organizational configurations of

$$\sigma_{AB}, \qquad R_{ABCD}, \qquad L_{AB}.$$

Internal degrees of freedom may be interpreted as internal organizational modes of stable vortices.

The current correspondence scheme is

$$V_P \longrightarrow \text{proton-like state},$$

$$V_N \longrightarrow \text{neutron-like state},$$

$$V_C \longrightarrow \text{composite bound state}.$$

Within Stage VIII, matter provides the principal bridge between continuum organization and observable physics.

The correspondence hierarchy may therefore be written as

$$V \longrightarrow M \longrightarrow \text{Observable Physics.}$$

A complete classification of stable particle-like vortex states remains a subject of future development.

19 Open Problems

Although MCM V08 provides a coherent ontological framework, several fundamental questions remain open.

These questions do not challenge the ontology of the model itself.

Rather, they identify areas requiring further mathematical and conceptual development.

1. What is the precise mathematical structure of

   $$\Sigma_G$$

   as the global organizational state of the medium?

   At present,

   $$\Sigma_G$$

   is defined ontologically.

   A complete mathematical description of global continuum organization remains to be developed.

2. What is the explicit mathematical form of

   $$c = F(\Sigma_G) \; ?$$

   The MCM identifies propagation as a consequence of global continuum organization.

   However, the functional relationship between

   $$\Sigma_G$$

   and

   $$c$$

   has not yet been established.

3. How does the local stability condition

   $$L_V \tau_V = \Lambda$$

   relate quantitatively to the global organizational state

   $$\Sigma_G?$$

   The connection between local bound organization and global continuum organization remains an open problem.

4. How do stable classes of bound geometry emerge?

   The conditions under which distinct stable vortex configurations form, persist, and transform require further investigation.

5. How should the emergence of observable matter be classified within the hierarchy of stable bound structures?

   A complete taxonomy of vortex-derived matter states has not yet been established.

6. Can the complete continuum hierarchy

$$D \longrightarrow \Sigma_G \longrightarrow \sigma_{AB} \longrightarrow \gamma_{AB} \longrightarrow R_{ABCD} \longrightarrow L_{AB} \longrightarrow V \longrightarrow L_V \longrightarrow \text{Space} \longrightarrow \text{Matter}$$

   be expressed as a fully closed mathematical system?

   This remains one of the central long-term objectives of the MCM research program.

These open problems define the next development stages of MCM V08.

Their resolution would transform the model from a coherent ontological framework into a fully formalized continuum theory while preserving its monistic foundation.

No additional ontological entities are required for this development.

All future extensions must remain fully contained within the ontology of

$$D.$$

20 Conclusion

MCM V08 establishes a monistic continuum ontology in which all observable structures emerge from a single continuum

$$D.$$

The theory introduces a global organizational state

$$\Sigma_G = \frac{S_G}{S_{G,0}},$$

which provides the foundation for both local organization and global propagation.

The emergence hierarchy

$$D \longrightarrow \Sigma_G \longrightarrow \sigma_{AB} \longrightarrow \gamma_{AB} \longrightarrow R_{ABCD} \longrightarrow L_{AB} \longrightarrow V \longrightarrow L_V \longrightarrow \text{Space} \longrightarrow \text{Matter}$$

establishes a coherent organizational framework.

The propagation hierarchy

$$D \longrightarrow \Sigma_G \longrightarrow c$$

and the matter correspondence hierarchy

$$D \longrightarrow \Sigma_G \longrightarrow V \longrightarrow M \longrightarrow \text{Observable Physics}$$

provide the connection between continuum organization and physical observables.

Local bound structures satisfy

$$L_V \tau_V = \Lambda,$$

whereas global propagation is determined by

$$c = c_0 \Sigma_G^n.$$

These relations describe different organizational levels of the same continuum.

Stage VIII establishes the correspondence between ontology and observable physics while preserving the strictly monistic character of the model.

All observable structures remain manifestations of the organization of

$$D.$$

With the completion of Stages I–IX, the current MCM development roadmap is considered complete.

References

[1] Moosbrugger, W. *Monistic Continuum Model (MCM): V07*. Zenodo, 2026. DOI: https://doi.org/10.5281/zenodo.22128163

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